

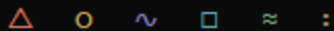


FINAL REVIEW

a sensorial-comfort copilot for architectural layouts

the whole instrument, end to end

Charles Abi Chahine · Emilie El Chidiac · Lakzhmy Zaro · María Sánchez Domínguez



We measure everything. Except how it feels.

Energy, cost, code: all measured. The lived sensory experience of a room is not. And the **ripple** between the senses, where every design move has a hidden second cost, is invisible.



change one, the others move

Sensi makes that invisible axis, and its trade-offs, measurable.

The edges, made visible.

A space is a conversation between your senses. The overall score is the least interesting part. The lesson lives in the edges, and we prove each idea where it shows.

PERSONAL

your senses, your weights. Not an average body. [proved at Onboard]

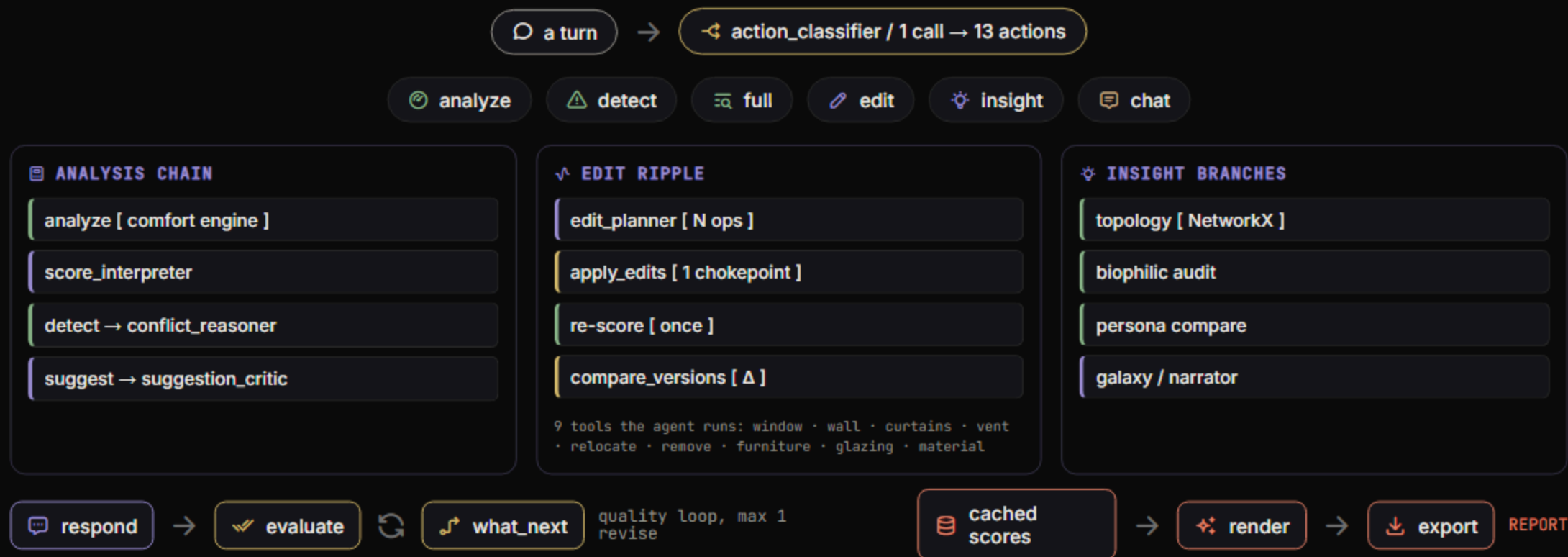
HONEST

no averaging. The worst sense gets a one-vote veto. [proved at Shape]

COUPLED

senses pull on each other. That pull is the ripple. [proved at Shape]

You talk to it, and it figures out the rest.



FastAPI / LangGraph / React · in-process comfort engine, no Rhino · streams its thinking, ~3s to first feedback

Three models, each for a different job

on the current **Gemini 3.x** generation, verified live against Google's docs before trust.

tier	model ID	does	\$ / 1M (in · out)	context
 FAST	gemini-3.1-flash-lite	routing / classification	\$0.25 · \$1.50	1.0M
 SMART	gemini-3.5-flash	user-facing reasoning	\$1.50 · \$9.00	1.0M
 IMAGE	gemini-3.1-flash-image	room renders (Nano Banana 2)	~\$0.067 / img	128k

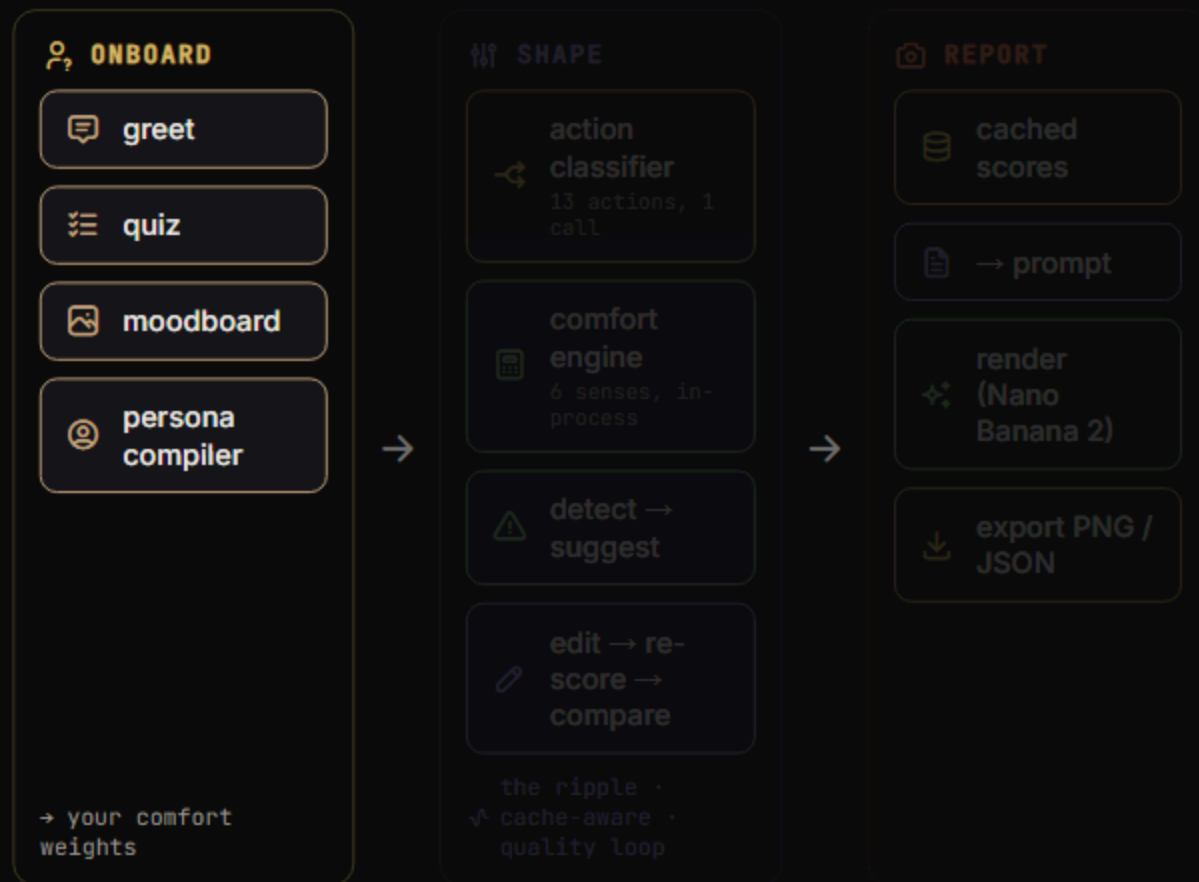
CONSIDERED, NOT ADOPTED
3.1-pro-preview top reasoning, ~8× cost, preview / nano-banana-pro 4K hero renders, ~3× img cost

WHY TWO TIERS
small models route the turn; the reasoning tier runs only where quality compounds.

You tell Sensi who you are

A quiz, then a curated moodboard. Sensi
compiles a sensory persona.

proves it's personal



▶ DEMO 1 / 3 · ONBOARD



CLIP 1 / ONBOARD

~50s · pre-recorded narrated demo

quiz → moodboard → the earned persona reveal · drop assets/clips/onboard.mp4

Your answers turn into your own scale

Signals from onboarding raise per-sense **weights** (they only ever go up):

q5 "plants" non-negotiable +0.18

q3 "noise bothers me" floor to 0.80

inspire "warm wood" +0.08

→ Emilie: thermal .90 / acoustic .80 / olfactory .90 ...

A WEIGHT SETS THAT SENSE'S ALERT THRESHOLD

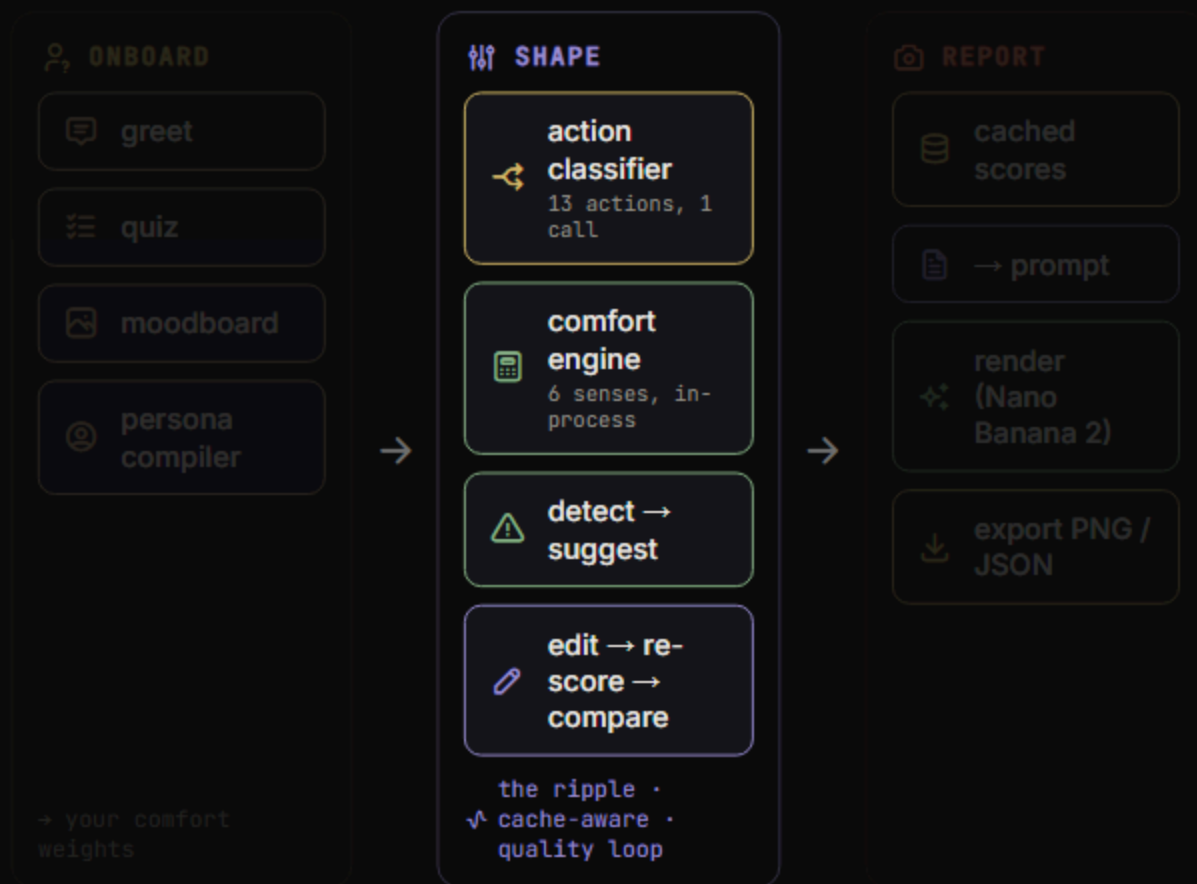


care more about a sense → Sensi flags it sooner. The reveal teaches the model the engine *actually* computes, from its own constants.

The conflicts you couldn't see

Score six senses on a real CAD plan, find the conflicts you can't, then edit and watch it ripple.

proves the honest number + the ripple





CLIP 2 / SHAPE

~90s · pre-recorded narrated demo

CAD plan → scores → conflict → edit → Focus Pull → re-score → ripple graph → galaxy ·
drop assets/clips/shape.mp4

See what the design does to each sense

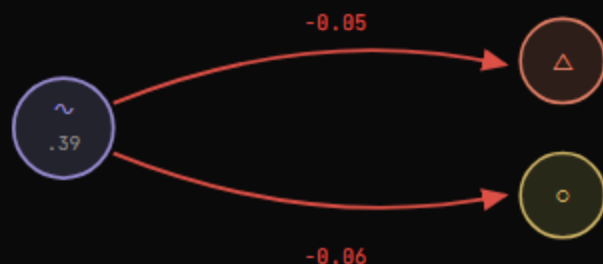
worked example **Kitchen / layout 201** → room-type baseline, then design levers from the real geometry:

sense	baseline	design levers (from the plan)	effective
△ thermal	0.50	N orientation -0.03	0.47
○ visual	0.70	glazing < 10% -0.15	0.55
~ acoustic	0.45	reverb on hard tile -0.06	0.39
□ spatial	0.60	volume / norm = 1.7 ↑	1.00
≈ olfactory	0.35	mech vent -0.05 · wet adj -0.10	0.20
: tactile	0.60	ceramic floor / plaster walls	0.55

every lever is research-grounded physics: orientation, glazing ratio, Sabine reverberation, ventilation, material absorption. next: the senses pull on each other
→

Senses pull on each other

A failing sense drags its partners. Kitchen acoustic **0.39** pulls down thermal and visual:



`src < 0.50 → partner -0.06 / src ≥ 0.70 → +0.04`
labelled with why: building science / physics

Then the worst sense vetoes

No averaging. Half the weighted mean, half the single worst sense:

```
overall = (1-v)·mean_w +  
v·worst
```

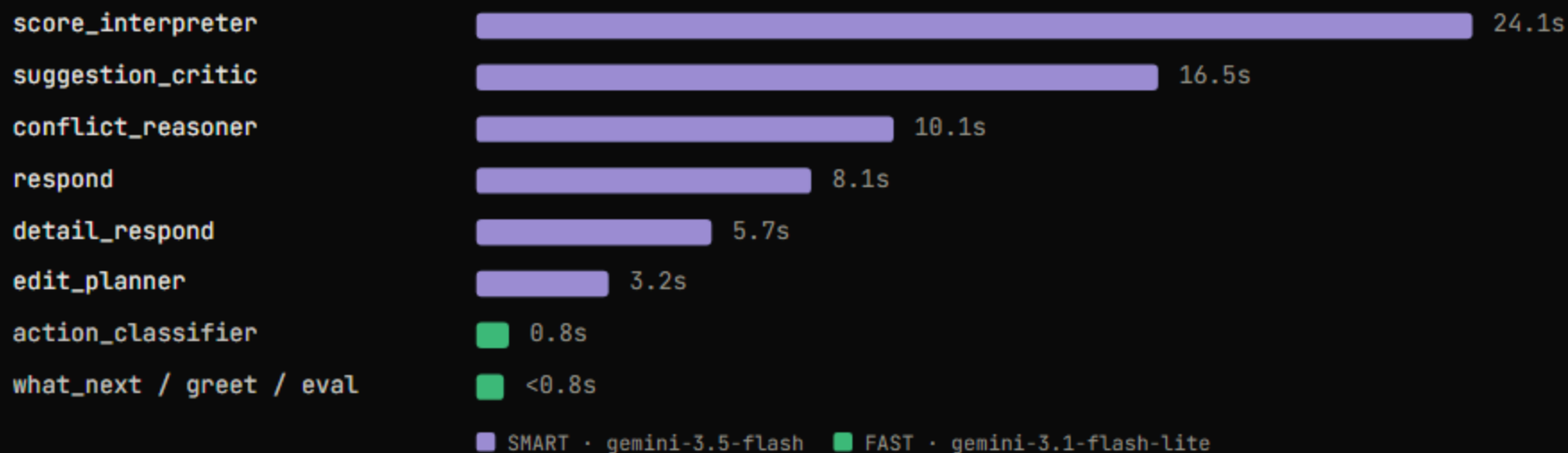
`v = 0.5 · sense_model.py`

0.49 **0.20** **0.34**
weighted mean + worst (olfactory) = Kitchen overall

a great room with one unbearable sense is still unbearable

Cheap models for the easy stuff

one scripted session, every node measured. FAST stays sub-second; SMART is where the time + cost live.



FULL SESSION

\$0.048

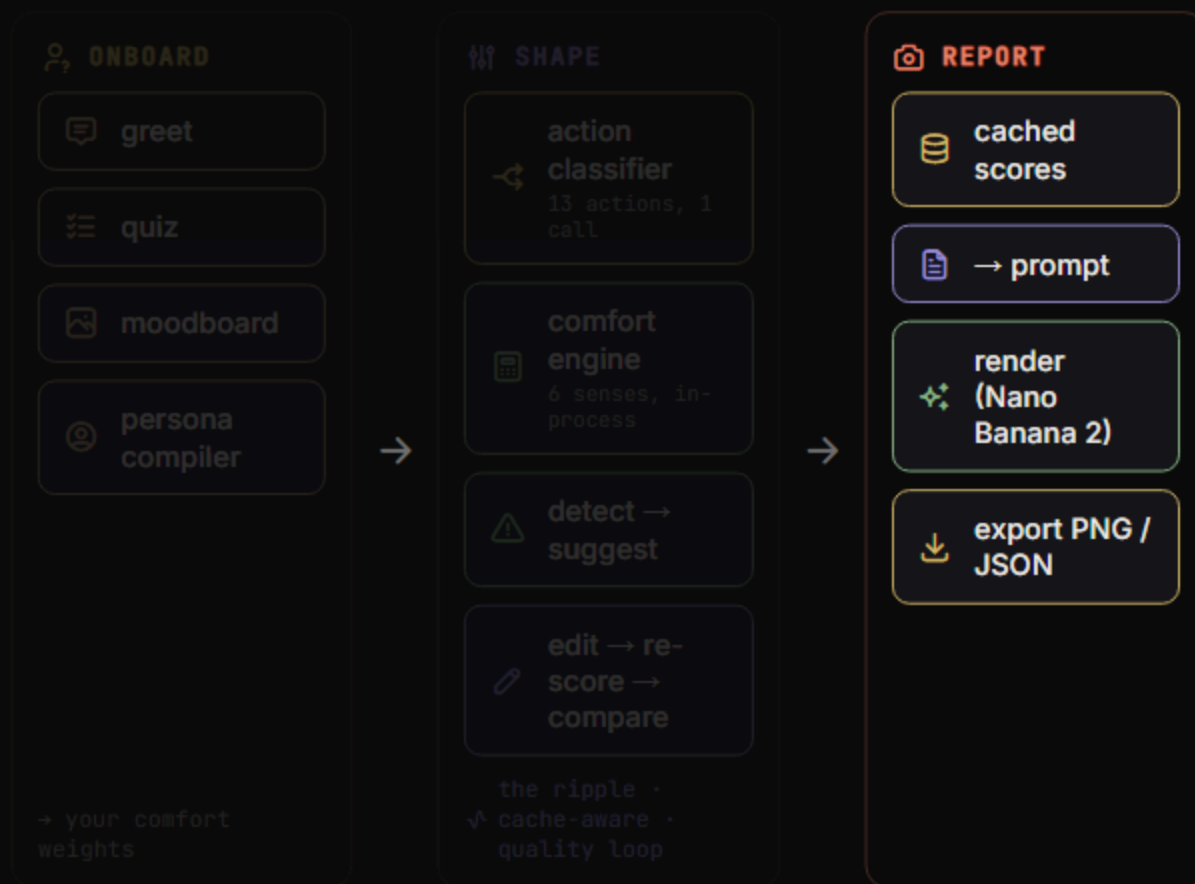
• under a nickel

vs the old 2.5 tier: SMART +33% slower, session +3.8× costlier (3.5 Flash's thinking budget). FAST stays near-free. **we pay latency + cents for capability.** Judged blind, the new reasoning still won 4 of 5 nodes: no regression.

A vision of how each room feels

The output answers the input. Your persona and aesthetic return, and every room becomes an image of how it feels.

proves made real





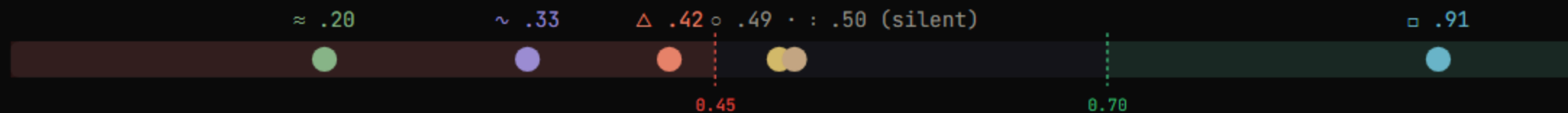
CLIP 3 / REPORT

~70s · pre-recorded narrated demo

persona + moodboard return → per-room renders → before/after · drop
assets/clips/report.mp4

Only the extremes make it into the picture

the Kitchen's scores on a 0→1 band. Below 0.45 or above 0.70 writes a phrase; the mid stays silent.



"...stuffy, closed air [≈]; hard reflective surfaces [~]; a cool, bluish feel [Δ]; open and generous in volume [□]."

deterministic build_room_prompt() → Nano Banana 2

input → output: the circle closes

BEFORE / AFTER · the same room, your edit applied



how it holds: the "after" is the room's canonical render; the "before" is that same image edited by a "what changed" clause, so the scene stays put and only the change moves.

We tried two image engines

GOOGLE vs OPENAI (gpt-image-1)



Google • 6.7s • \$0.039



OpenAI • 20.5s • \$0.042

Google: clean architectural daylight. OpenAI: darker, cinematic. → we stay on Google: $\approx 3\times$ faster, a touch cheaper, steadier across edits.

NEW IMAGE MODEL vs OLD, judged blind

3 / 3

rooms, the new
model
judged sharper

higher-resolution, photorealistic light vs the old model's flatter CG. **how it feels, made believable.**

It's grounded in research

The couplings come from peer-reviewed building science and standard room physics. Every research claim went through an adversarial fact-check; the ones that didn't survive were cut. Where evidence is thin, we encode the *direction*, not a fake-precise number.

R1 autism sensory taxonomy / R2 multisensory mind / R3 cross-modal thermal / S2 Yang & Moon 2019 / S6 ASHRAE

THE COUPLING MATRIX · how senses move each other

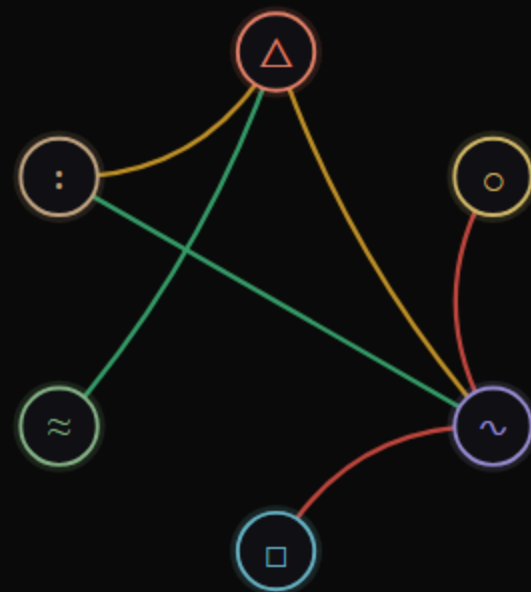
	△	○	~	□	≈	:
△			±		+	±
○			·			
~	±	-		-		+
□			-			
≈	+					
:	±		+			

+ helps - harms ± trade-off · one-way (no reverse effect)
solid = research-verified dashed = standard physics

18 / 25 couplings kept by a 3-vote adversarial test · direction over false precision.

The score was never the point. The ripple was.

the first tool that makes the **coupling between senses** designable,
not just measurable.



six senses, and how they pull on each other

we measure everything except how it feels → now we can argue about it.

What we're thinking about

LEGIBLE vs OPTIMAL

Should comfort be *optimized*, or just made *legible* so a designer decides?

WHOSE COMFORT

When a household disagrees, whose senses win? Who sets the weights?

PROPOSE vs DECIDE

How far should the agent go: surface the trade-off, or make the call?

And the honest one: is a felt-experience score accountable, or just persuasive?



thank you

now let the senses talk

△ ○ ~ □ ≈ :

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